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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CST) (CBCS) Degree Examinations
 May – June 2025

22CST223 – DESIGN OF EARTHQUAKE RESISTANT STRUCTURES

Duration: 3 Hours

Max. Marks: 100

Note: Answer *Five full* questions choosing *Two full* questions from **Unit – I & Unit – II** each and *One full* question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write a Note on elastic rebound theory and continental drift theory.	10	L*2	1	1
	b) Explain Characteristics of Earthquake and its quantification.	10	L2	1	1
2.	a) With a neat sketch explain the types of Seismic measuring instruments.	10	L2	1	1
	b) Discuss Seismic Zone map of India as per IS -1893-2016.	10	L4	1	1
3.	a) Explain the following				
	i) Elastic and Elasto-plastic spectra	10	L2	1	1
	ii) Tripartite Plot	10	L2	1	1
	b) What are surface waves? Explain its types.				
		10	L2	1	1
Unit – II					
4.	Discuss Key points of IS 1893: 2016 Part 1.	20	L4	2	2
5.	A Plan and Elevation of a Three story RCC School building is shown in figure 5. The building is located in seismic zone V. the type of soil encountered is medium stiff and its proposed to design the building with a special moment resisting frame. The intensity of DL is 10kN/m ² and the floor are to cater to an IL of 3kN/m ² . Determine the seismic design loads on the structures by static analysis				

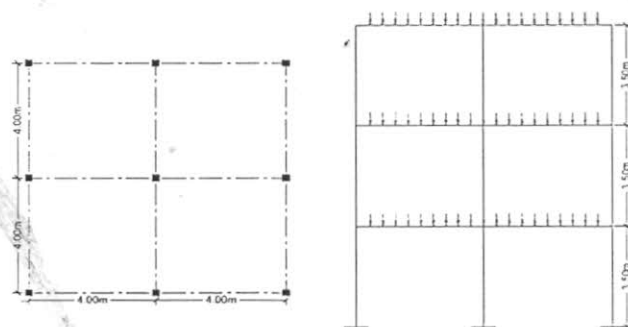


Fig. Q 5

6.	a) Write a note on Base isolation.	20	L5	2	2
	b) How Earthquake will affect the RC building?	10	L2	3	3
		10	L4	3	3
Unit – III					
7.	a) Discuss Earthquake resistance design of masonry building.	10	L2	4	4
	b) Write a note on rigid diaphragm with neat sketch.	10	L2	5	5
8.	a) Explain causes of failure of masonry structures and what are remedial measures taken to retrofit the structures?	10	L2	5	5
	b) Discuss seismic behavior of masonry buildings during earthquakes.	10	L4	4	4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
